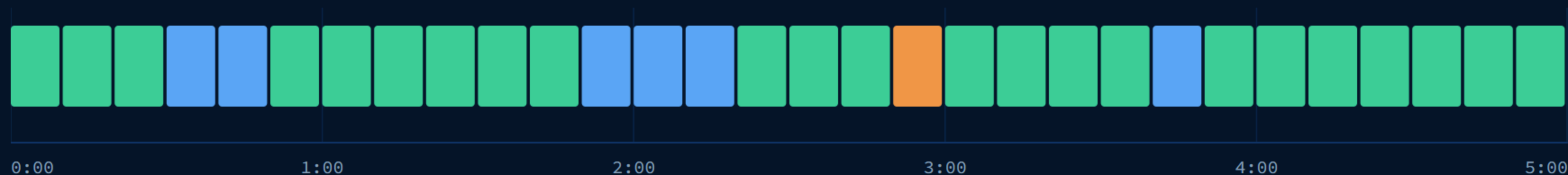


# Your Vision Model Is Biased

Lessons from shipping real-time video detection

Serban Petrescu • Trilogy / TimeBack • Compass AI 2026



# No teacher in the room, and no one watching the test

## WHAT WE BUILD

- Trilogy's new products division: the software behind **Alpha School**
- A private US school: digital-first, AI-first, physical and online campuses
- No teachers: adaptive learning apps, and guides in the room
- Progress checked with periodic standardized tests

## THE PROBLEM

- Students alone with devices **cheat**, idle, and work on the wrong thing
- Official third-party tests kept bringing surprises
- The answer: proctoring software

## WE DETECT

- Cheating
- Wasted time
- Self-harm behaviors, raised to a safety officer

# One billion students. Zero humans.

- The founder's goal: improve education for **1 billion students** in 20 years.
- Humans bootstrap the detectors. Every human left in the loop is one we are working to remove.
- Today, no human reviews a production detection before the student sees it.

**1**

engineer built it

**2 mo**

to real student traffic

**140k**

hours of video a month

**3,400**

students, last 30 days

**335B**

tokens a month

**Eighteen months of mistakes. Every one was a bias we put there ourselves.**

# Three ways students cheat

## UNAUTHORIZED DEVICE USE



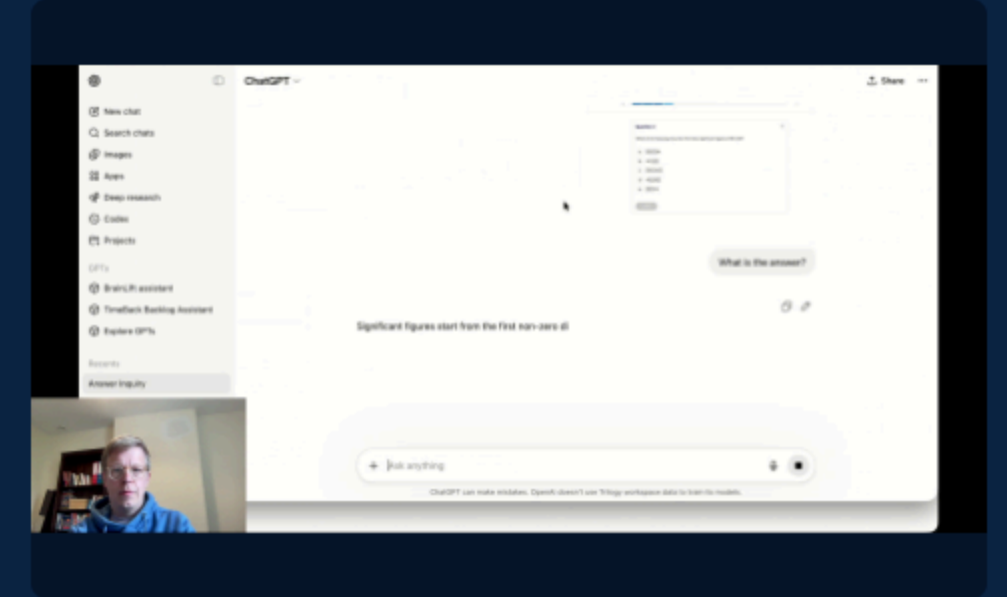
A handheld device, visibly in use. Judged from the webcam alone.

## HELP FROM ANOTHER PERSON

***“The answer is B.”***

Another person supplies the answer to the active question.

## UNAUTHORIZED APP USE



An external tool used to obtain the answer to the question on screen.

# Every flagged behavior has an allowed twin

## FLAGGED

### Device use

Phone in hand

### Help from another person

“The answer is B.”

### App use

Pastes the question into ChatGPT, submits the answer it gives

## ALLOWED

Phone lying on the desk, untouched

“Your screen froze, let me fix it.”

A calculator, on a math test that allows one

**To the camera, each pair looks nearly identical.**

Boundary rules from the production annotation instructions; the same rules humans are graded against.

# Precision below 0.90 disqualifies a detector

0.90

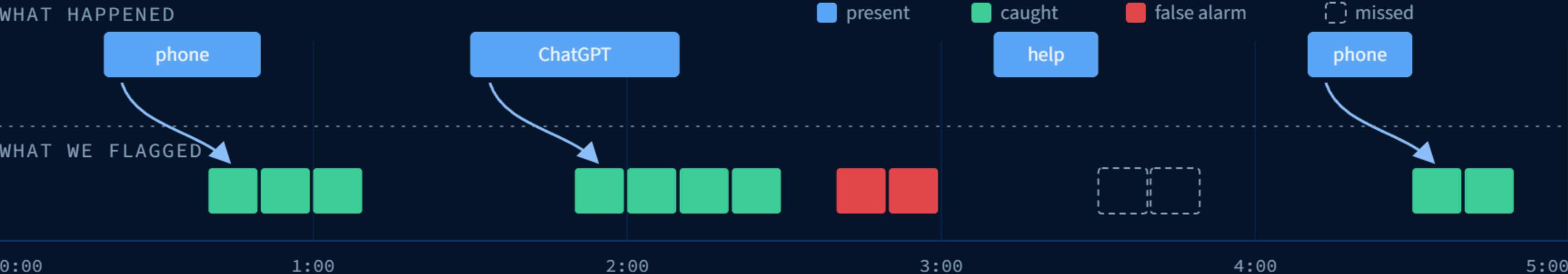
precision: of what we flag, the share that is real.  
0.95 is the goal. The hard gate.

0.90

recall: of what happened, the share we catch.  
Second priority.

≤2 min

latency, event to warning. Constant, never accumulating.

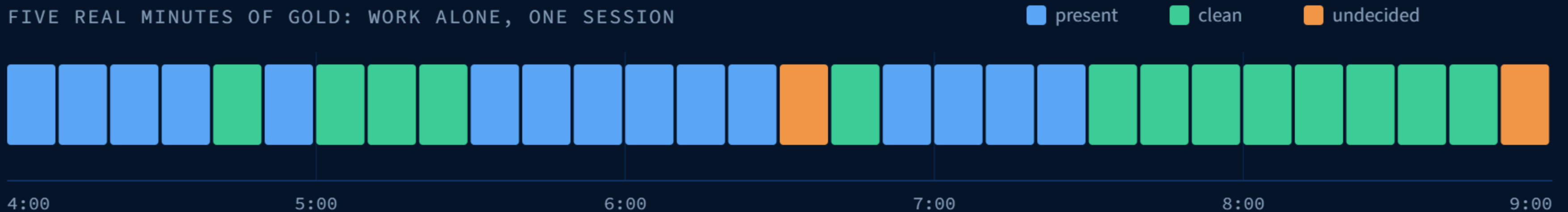


No single flag ever punishes a student on its own.

# We build gold ten seconds at a time

- Every week brought a new model and a new idea. Every demo looked good.
- Picking needs a harness: run every idea against the same sessions, scored the same way.
- Pre-trained or not: a quality claim needs a figure behind it.

FIVE REAL MINUTES OF GOLD: WORK ALONE, ONE SESSION

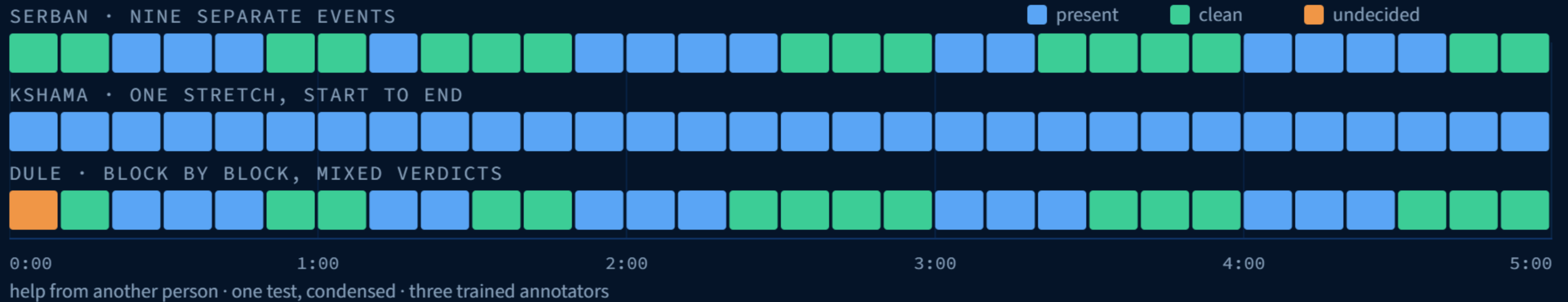


every block carries polarity, difficulty and a written reason. One of the orange blocks:

*“The student observes something off-camera and speaks, maybe to herself. Surely about the test question though.”*

**We annotate in 10-second blocks to keep it simple. In around 13% of them, humans could not decide on the first pass.**

# Six annotators, one session, three verdicts on the same ten seconds



The same ten seconds. Opposite verdicts, all with confident reasons:

SERBAN **FLAGGED**

*“Done fully by the peer. No input from the student.”*

KSHAMA **FLAGGED**

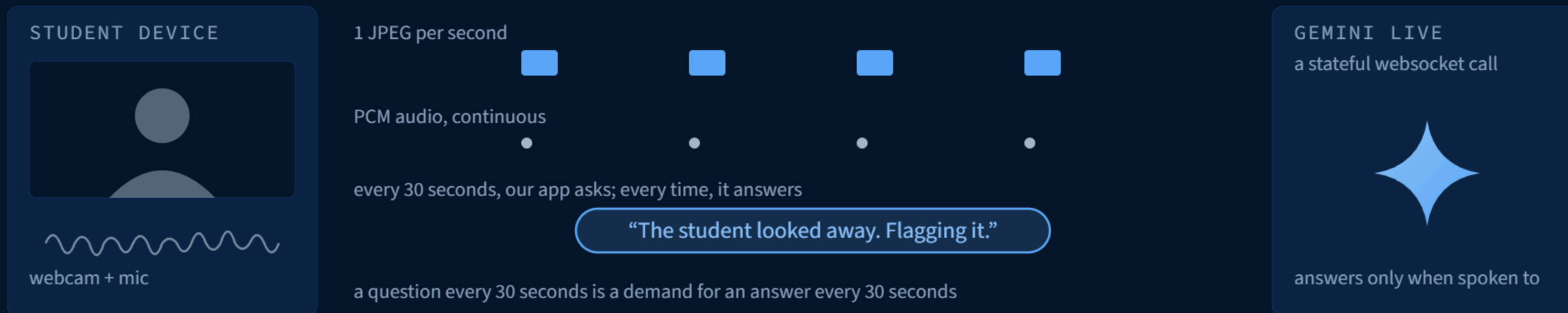
*“An older sibling sits next to the student for the entire test.”*

DULE **CLEAN**

*“Three children are present, but nothing contributes to an answer.”*

Disagreement was the point: every divergence told us which rule to write.

# Attempt 1: a model that only speaks when spoken to



**n/a**

precision: never measured.

**n/a**

recall: never measured.

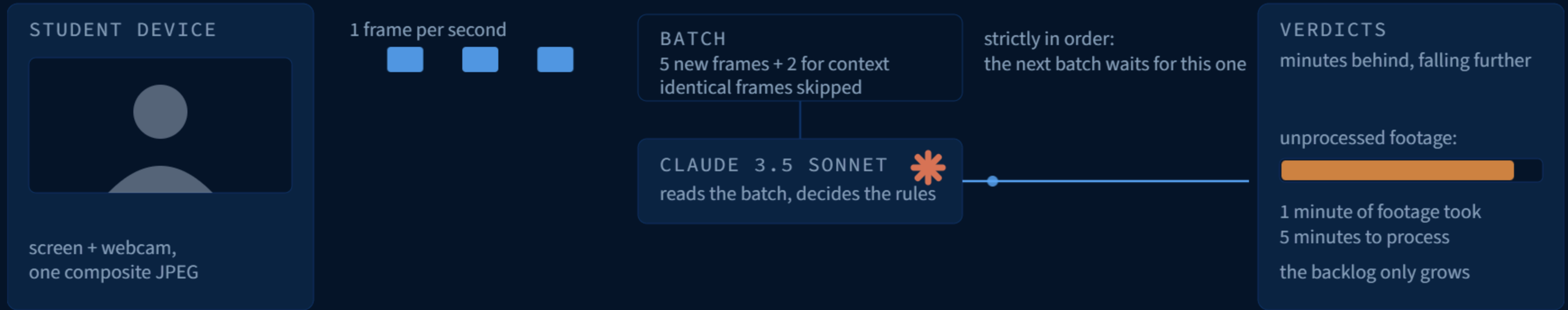
**30 s**

latency: the nudge cadence, by construction.

**Ask a model every 30 seconds whether anything happened, and it will tell you something did.**

Checked September 2026: Live sessions now run unlimited, the model may decline to answer, and input is still one JPEG per second. Replaying a live stream against gold is still impractical.

## Attempt 2: stills gave 0.20 recall and a spiralling backlog



# 0.70

precision: far below the 0.90 gate.

# 0.20

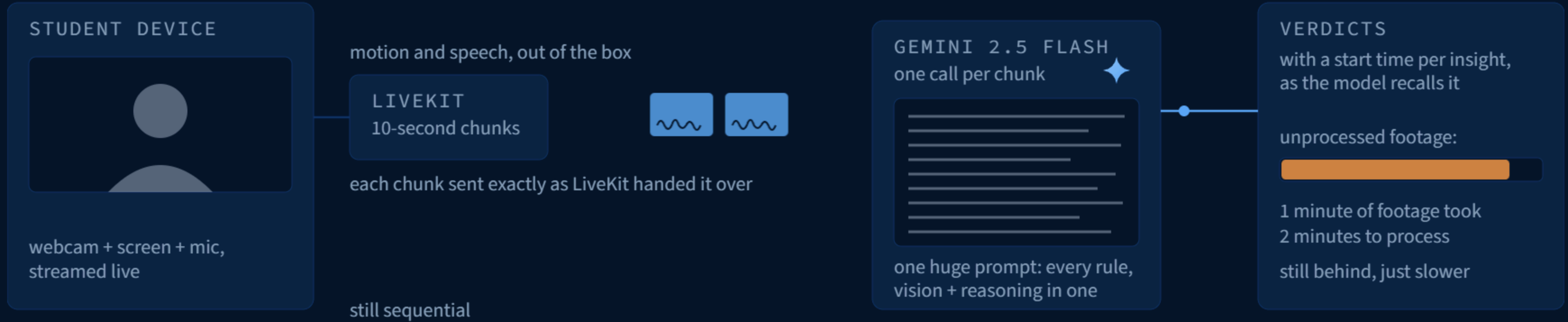
recall: stills are blind to motion and deaf to audio.

# 5x

latency: a backlog that compounds all session.

## Blind to motion, deaf to audio, and \$1,000 per student per month.

# Attempt 3: audio doubled recall and hid the precision gap



## 0.80

precision, on our annotated sessions. So we thought.

## 0.40

recall: audio and motion doubled it.

## 2x

latency: still compounding.

## We shipped it everywhere, and our own number was wrong by a factor of eight.

In production July 2025 to August 2026: 72,000 hours of footage, on a few hundred students for most of that span.

# “They are drinking from the phone”



## PROCTORING INSIGHT

type: UnauthorizedDeviceUse  
confidence: **HIGH**  
duration: 8 sec

*“Student is holding a phone in their hand and appears to be **drinking from it** while the assessment is active.”*

# 90%

of flags reaching human review were false.

# 8

sessions in five weeks, same behavior, same flag.

# HIGH

confidence, on every one of them.

## No human would ever say a student is drinking from their phone.

Production human-review queue, June 2026: eight sessions, 8 April to 11 May 2026, each one a water bottle flagged as a phone. Reproduced in isolation on a three-line prompt. Photo: Markus, re-enacting it.

# Prompting harder made it worse

FALSE ALARMS ON THE THREE HARDEST SESSIONS, BY QUESTION SHAPE PRECISION, PROMPT VERSION BY PROMPT VERSION

“Is this person holding a phone?”

41% false

the production prompt: 400 lines of rules and exclusions

33%

“What object, if any, is in each hand?”

27%

describe first, classify the text after

27%



and the input mattered four times more: the webcam was 10% of the pixels we sent • 67% → 40% on that change alone

**Every prompt line fitted the prompt harder to the videos we had handpicked.**

# Making the pipeline dumber is what raised precision

CUT THE SESSION INTO WINDOWS. ASK EACH ONE QUESTION AT A TIME.



## THE VISION PIPELINE, KEPT DUMB

one window, one rule, yes or no

stateless, parallel, replayable

one question per call

## DETERMINISTIC CODE

When did the test start and end?

Is a calculator allowed right now?

**no model at all**

## ONE SMART MODEL, AT THE END

reads all the text, builds the story of the test

reads text only, never sees a frame

after the test, not during

August 2026 cutover. Wherever a hard signal existed (test state, full-screen, window focus), vision stopped being consulted at all.

# Separating the eyes from the judge cleared the gate

60 s window



## Narrator

a person seated at a desk facing the screen

the screen shows a multiple-choice question

a dark rectangular handheld object raised to face height, screen side toward the person

no second person, no audible speech

## Classifier

device in use? **yes**

someone helping? **no**

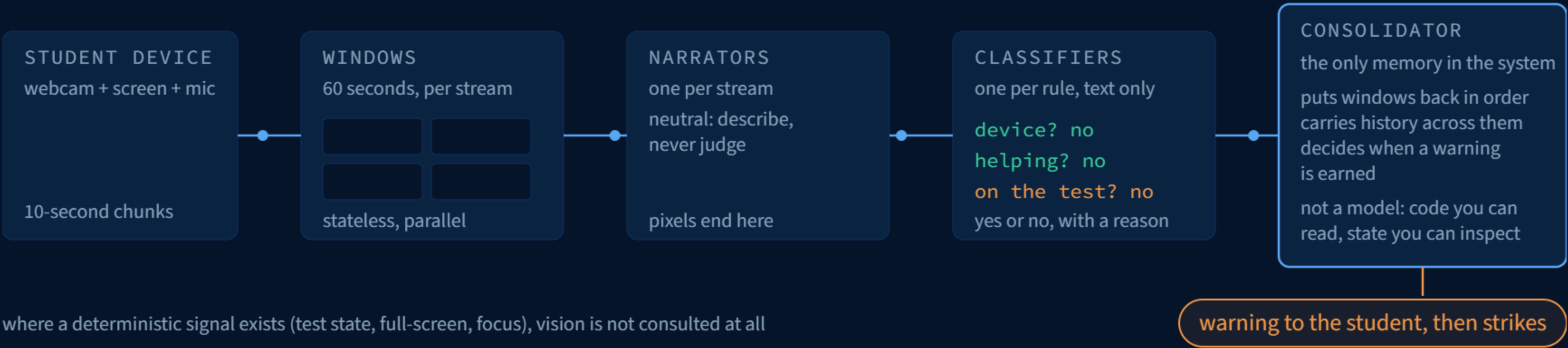
only the test on screen? **yes**

face in view? **yes**

**The model that decides never sees a pixel.**

Khayatan et al. 2026, When Prompts Override Vision: a question that presupposes an object is the largest single cause of hallucination, and describe-then-classify is the structural fix. We had shipped the same split before we read it.

# The window graph holds 0.95 precision at a constant two minutes



0.95

precision on live traffic, against gold and human review.

0.90

recall, measured the same way.

2 min

from the event to the warning. Constant.

Stateless everywhere, except one place built to remember.

In production since the August 2026 cutover: 145,000 hours of footage. A detector reaches this rung by clearing 0.90 on the gold set, then 0.90 precision across 200 human reviews.

# 62 models auditioned for one job

THE FUNNEL, RE-RUN EVERY MONDAY

62 models watched across every vendor catalog

can watch a full minute of video

most cap at 5 images

allowed to see student video

US-hosted, DPA

baked against prod

same run, same sessions, 3 reps

above the floor

precision 0.90, at a carryable cost

1 job

**THE CHAMPION • gemini-3.1-flash-lite**

nobody beats it on precision, recall, cost and latency together

HOW AUDITIONS END

glm-5.3-flash **extremely high error rate**

pegasus-1.5 **quotas don't even support our tests**

nova-2-lite **never returns the schema it was asked for**

nova-pro **fires on half the hard tail**

gemma-4 **6.5 minutes to watch 1 minute**

claude-sonnet-5 **57x the cost**

gemini-3.8-flash **ties on quality, 2.4x the cost**

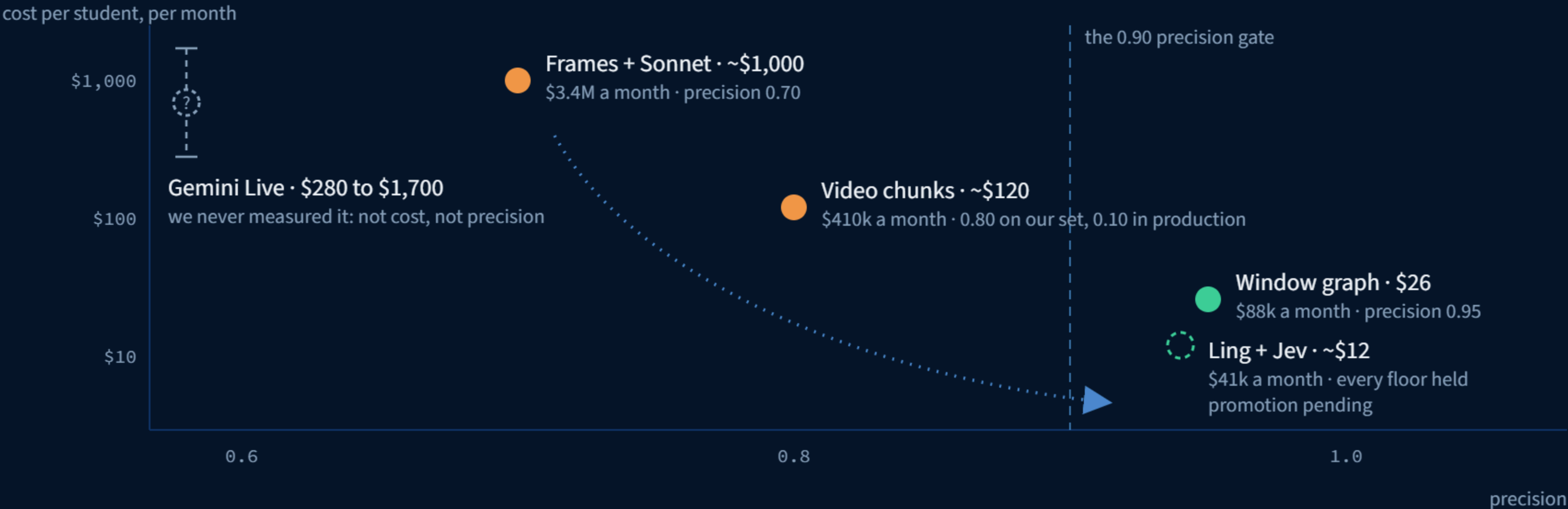
**THE CHALLENGER, CONFIRMED THIS WEEK • Ling + TypeSafe Jev**

holds every precision floor at half the cost; the promotion call is pending

## Every new model gets a blind check first: what do you see? what do you hear?

The standing challenge: catch two more true windows than production on one insight, hold precision at 0.90, at a cost we can carry.

# Cheaper and better are the same direction



The cheapest system we ever ran is the most accurate one. That is not a coincidence.

Vision model only, list prices, against a fixed unit: the normalized student at 34 recorded hours per month. Monthly totals at today's scale.

# Six rules we would follow if we started again

1. Build gold before you build anything: annotate to align, and let the disagreements write the rules. BIAS 1
2. Set the bar before the architecture: precision is a gate, recall a target, latency a constant.
3. Keep it simple: no silver bullets, however strong the launch benchmarks looked. BIAS 2
4. The question shapes the answer: keep rule words away from the eyes, and your own bias out of the dataset. BIAS 5
5. Make the pipeline dumber: deterministic where code can decide, one binary question where it cannot. BIAS 4
6. Measure the input before you judge the model: check what it actually receives, and what it perceives. BIAS 3 BIAS 6

**Every mistake was a bias we put there ourselves. That is the good news: so were all the fixes.**



# Thank you

Questions?

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